

GEOS-MPM Verification Suite Report

Generated by verification/runSuite

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1 Running and rerunning

Cases are staged under `testRunDirectory/verification/<case>`. The suite records failures and continues unless `--stop-on-error` is supplied. Use `--force` for noninteractive reruns.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter
verification/runSuite run --python /usr/tce/bin/python3 --force --jobs 4 --case-jobs 2
verification/runSuite report --python /usr/tce/bin/python3
```

2 Timing and dispatch summary

Case	Status	Family	GEOS job(s)	GEOS state	GEOS wall	Issue
surfaceInformedPolymerLayer	passed	surfaceInformedPolymerLayer	6290987, 6290987, 6290987, 6290987		dispatch 4.942, 2.243, 1.882, 1.866 s	

Subcase timing detail. Dispatch time is local wall time spent staging inputs, running PFW, and submitting the GEOS job. GEOS wall and CPU columns are populated from **sacct** when available.

Case	Subcase	GEOS job	State	GEOS wall	Total CPU	NCPUS	Dispatch
surfaceInformedPolymerLayer	Continuum tension	6290987					4.942
surfaceInformedPolymerLayer	Continuum compression	6290987					2.243
surfaceInformedPolymerLayer	Cohesive-zone tension	6290987					1.882
surfaceInformedPolymerLayer	Cohesive-zone compression	6290987					1.866

3 Case studies

3.1 surfaceInformedPolymerLayer

3.2 Surface-informed polymer layer

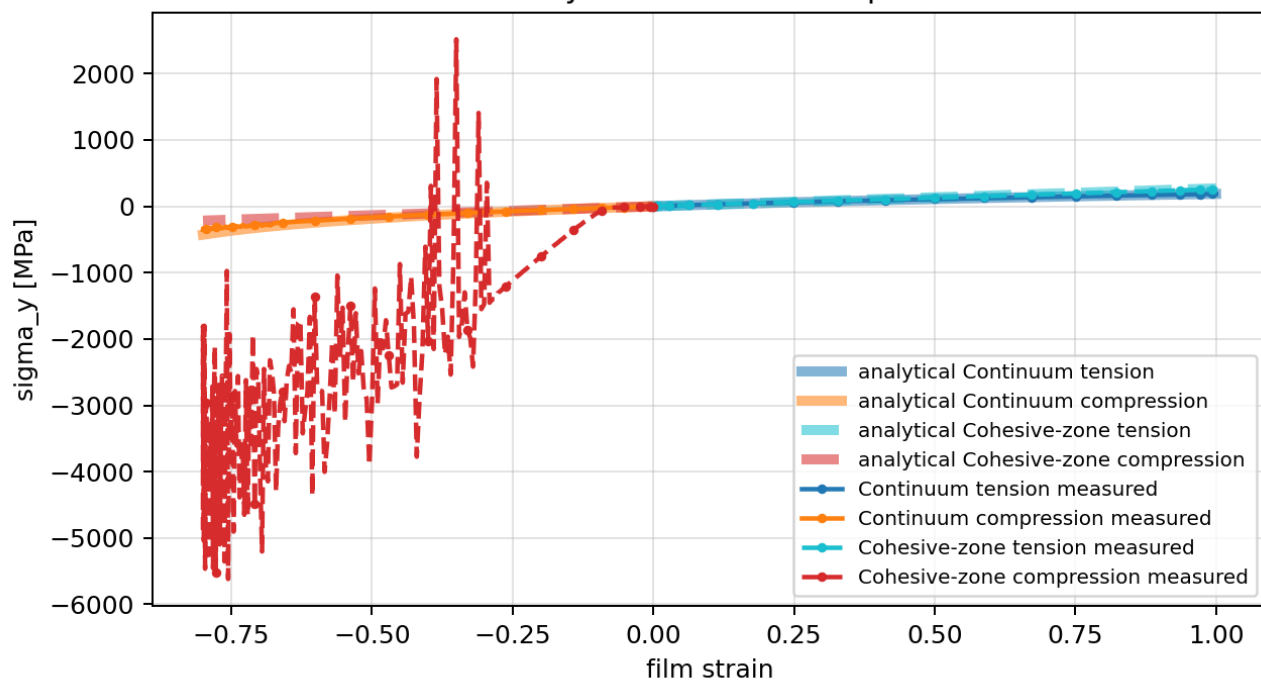
This verification folder exercises the surface-informed polymer as both a continuum material and a finite-thickness cohesive-zone material. Four variants are run from one input: continuum tension, continuum compression, cohesive-zone tension, and cohesive-zone compression. The continuum variants place a continuum polymer layer between stiff elastic loading bars and tie the two bar/polymer interfaces with stiff linear elastic cohesive zones. Those tie zones are not polymer material models; they are numerical bonded-interface fixtures that transmit load without relying on mixed continuum cells at the bar/polymer material interfaces. The cohesive-zone variants retain two stiff elastic blocks separated by a finite-thickness polymer cohesive interface with the same film thickness. The imposed cosine ramp is intentionally slow compared with the layer acoustic-transit time so the history is a quasi-static constitutive check rather than a wave-propagation benchmark.

The comparison curves use the continuum J2 radial-return update with retained mean stress for continuum variants and the reduced cohesive thin-film update with the same retained volumetric normal stress for cohesive-zone variants. In the cohesive projection, the mean film stress $p = K\epsilon_n$ is retained and only the deviatoric normal and shear stresses are returned to the scalar flow surface. The verification uses reusable non-polymer `ElasticIsotropic` block and `UncoupledCohesiveZone` tie cards from `pfw_materials.py`; the polymer response is carried by the continuum polymer layer in the continuum variants and by the polymer cohesive law in the cohesive-zone variants. The input explicitly requests `particleStress`, `particleDamage`, and `particlePlasticStrain` through `plottableFields`. The equivalent-plastic-strain visualization is a VisIt/post-processing expression computed from the primitive plastic-strain tensor components rather than a derived solver field. For measured histories, the film-strain coordinate is reconstructed from the actual deformation-gradient or length history written by GEOS when available, with the input cosine loading table used only as a fallback. When VisIt is available, it also renders loading-direction stress and equivalent plastic strain at four output states for each variant.

Quantitative result. The continuum variants place a finite-thickness continuum polymer layer between elastic loading bars; the cohesive-zone variants place a finite-thickness polymer CZ between elastic blocks. Measured curves use boundary reactions when present and fall back to material-volume-corrected box averages otherwise. The analytical comparison uses the GEOS material card constants; the continuum expected curve integrates log film strain from the finite-strain update, while the cohesive expected curve uses normal jump divided by thickness as in the CZ law. Cases that do not reach the target film strain are marked incomplete and excluded from error norms. Stresses are reported in MPa.

Variant	status	stress source	reaction measure	samples	final strain	target strain	RMS stress error	final stress
Continuum tension	complete	reactionHistory	internal_force	3e+02	1	1	1.01615	186.451
Continuum compression	complete	reactionHistory	internal_force	3e+02	-0.8	-0.8	35.8885	-353.746
Cohesive-zone tension	complete	reactionHistory	internal_force	3e+02	1	1	7.50469	256.713
Cohesive-zone compression	complete	reactionHistory	internal_force	3e+02	-0.8	-0.8	2210.85	-4874.72

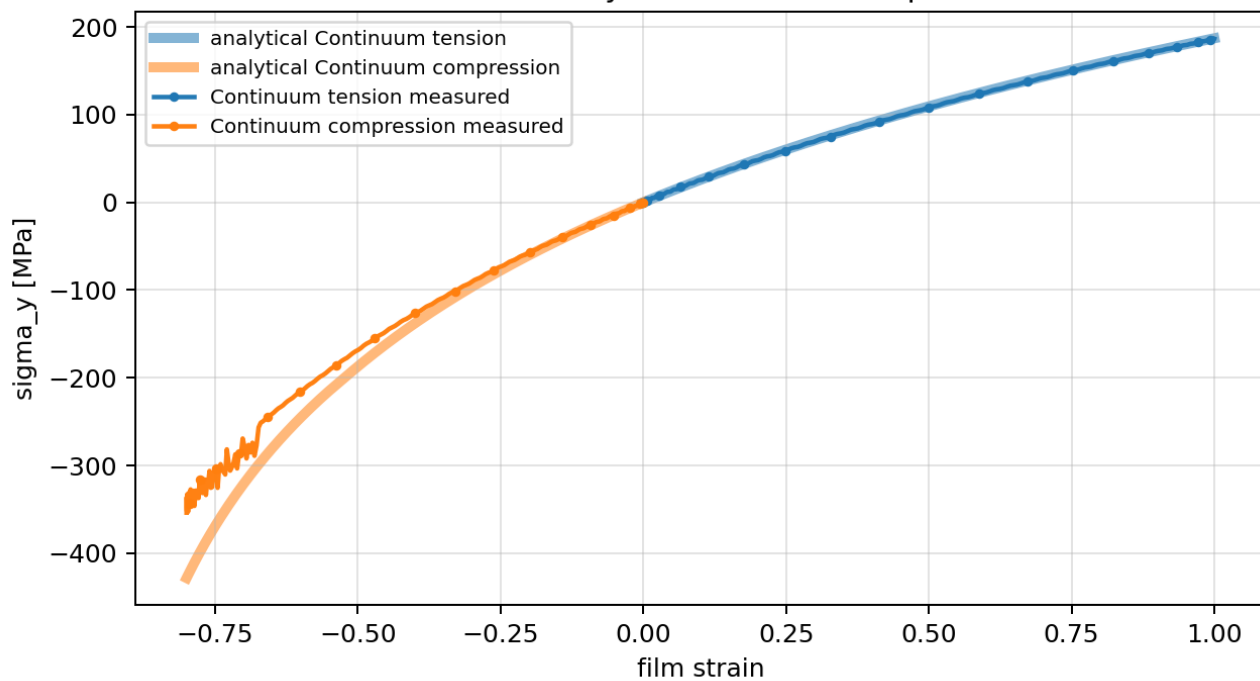
Thin-layer stress-strain response



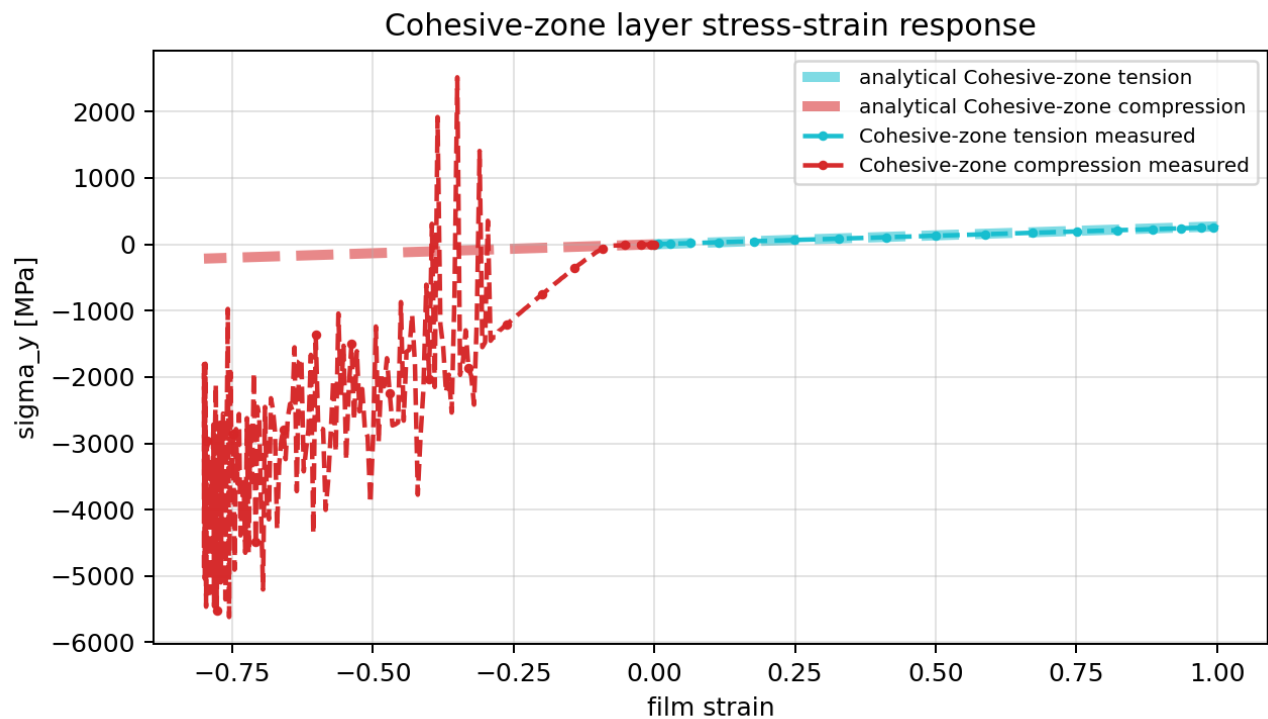
variants: stress-strain response.

All

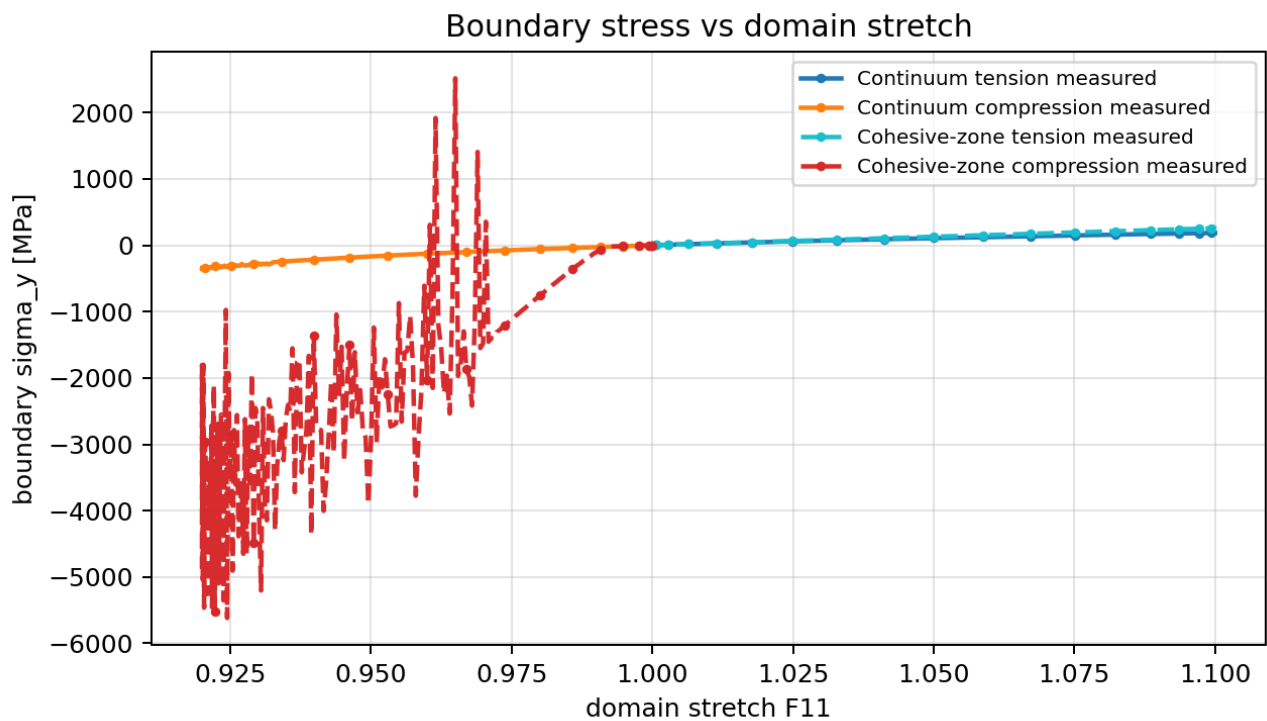
Continuum layer stress-strain response



Continuum variants: stress-strain response.

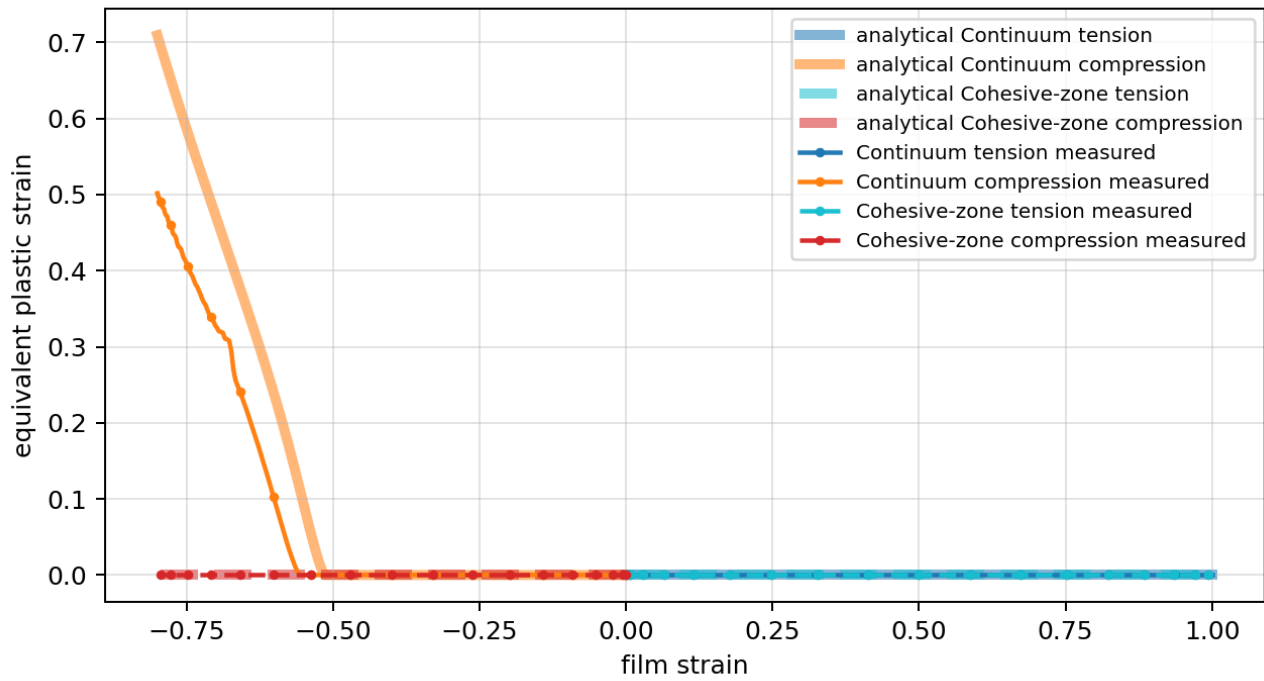


Cohesive-zone variants: stress-strain response.



boundary stress versus imposed domain stretch; this plot uses F_{11} directly and does not convert the x-axis to film strain. Measured

Equivalent plastic strain



plastic strain histories.

Equivalent